

Daniel Fernandez

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EDUCATION

Queens College, City University of New York (CUNY)

Expected Dec 2027

B.S. Computer Science — GPA: 3.47

Flushing, NY

- Relevant Coursework: Cloud Computing, Operating Systems (C++/C), Databases (SQL), Data Structures & Algorithms (Java), Advanced Python Programming, Software Engineering, Object-Oriented Programming (Java)

EXPERIENCE

Software Engineer Intern | Dow Jones & Company

Jun 2026 – Aug 2026

New York, NY

- Built an AI testing harness using dependency graphs to evaluate PR blast radiuses, automatically authoring Jest unit tests for modified components and executing targeted Playwright E2E runs via MCP on affected routes to prevent visual regressions.
- Diagnosed a visual and behavioral regression in a core WSJ module caused by missing props in Next.js SSR data-fetching, delivering a working code fix and technical RFC for senior engineering integration into platform infrastructure.
- Resolved an SEO compliance defect across 3 site sections serving 4.1M+ subscribers by fixing a shared component library bug that injected duplicate h1 tags into the semantic DOM hierarchy.
- Delivered time-sensitive homepage and news bucket configuration changes on WSJ's production site, partnering with product to turn around layout updates on tight deadlines.

Software Engineer Fellow | CUNY Tech Prep

Jul 2025 – Jun 2026

New York, NY

- Architected the PostgreSQL database for M.A.P.S., a full-stack academic planning platform built for 2,000+ students, writing complex SQL joins and transactions to structure how course schedules and professor metrics were stored and accessed.
- Unblocked full-stack development for the 4-person team by engineering a Puppeteer scraping pipeline to ingest real-time course catalogs, exposing the data via 10+ Express REST API endpoints secured with Firebase Authentication.
- Directed the technical execution of the platform (React, Node.js), leading critical architectural decisions for data modification workflows and conducting code reviews to ensure scalable delivery of the core application.

PROJECTS

codejam | *Java, Spring Boot, Node.js, Docker, React, TypeScript, WebSockets*

Aug 2026 – Present | [GitHub](#)

- Engineering a real-time collaborative code editor with zero document conflicts by architecting a split backend, utilizing Node for native JavaScript Yjs CRDT synchronization and Spring Boot (Java) for durable persistence.
- Optimized PostgreSQL database load for live documents by designing a write-back persistence model that uses shared in-flight promises to deduplicate cache misses and debounce timers for state flushing.
- Secured untrusted code execution while capping worst-case server compute by orchestrating network-isolated, memory-capped Docker sandboxes managed through a bounded Java thread pool.
- Delivered anonymous session tracking for collaborative code workspaces using server-issued HttpOnly cookies and idempotent database upserts, allowing users to retain history without forced authentication.

codeburrow | *Python, tree-sitter, Git, VoyageAI, ChromaDB*

Jul 2026 – Aug 2026 | [GitHub](#)

- Built a pip-installable hybrid code search CLI that improves retrieval relevance over standard baselines by fusing BM25 keyword search and Voyage AI embeddings via Reciprocal Rank Fusion.
- Increased embedding precision by designing a tree-sitter AST parsing pipeline that chunks multi-language source code (Python, JavaScript, TypeScript) strictly at the function and class levels.
- Eliminated redundant indexing overhead by implementing a git-aware post-commit hook that incrementally re-embeds only modified files while safely extending existing repository hooks.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Java, Python, SQL (Postgres), C++, HTML/CSS

Frameworks & Libraries: Node.js, Spring Boot, Next.js, React.js, Express.js, GraphQL, Playwright, Tailwind

Developer Tools: Git, Claude Code, Docker, Figma, Microsoft Office Suite

Infrastructure/OS: Linux/Unix